

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule for AI FAQ Assistant API. Each sprint carries 20 story points, for a total of 80 story points across four sprints. Team member entries are roles; replace them with names.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the NestJS project with environment config, PostgreSQL (with pgvector) and Docker so the team can build on a common base.	3	High	Backend Lead
Sprint-1	Account & Access	USN-2	As an admin, I can register my organisation's account by entering email, password and confirm password.	2	High	Backend Dev 1
Sprint-1		USN-3	As an admin, I will receive a confirmation email once I have registered.	1	Medium	Backend Dev 1
Sprint-1		USN-4	As an admin, I can log in with email and password and receive JWT access and refresh tokens.	3	High	Backend Dev 1
Sprint-1		USN-5	As an admin, I can assign roles (Admin, Editor, Agent) and the API restricts endpoints by role.	3	High	Backend Lead
Sprint-1	FAQ Management	USN-6	As an editor, I can create an FAQ with a question, answer and category (saved as a draft).	3	High	Backend Dev 2
Sprint-1		USN-7	As an editor, I can view a list of FAQs and open a single FAQ.	3	High	Backend Dev 2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1		USN-8	As an editor, I can update or delete an FAQ.	2	High	Backend Dev 2
Sprint-2		USN-9	As an editor, I can move an FAQ through Draft, Published and Archived states.	2	High	Backend Dev 2
Sprint-2		USN-10	As an editor, I can organise FAQs with categories and tags.	3	Medium	Backend Dev 2
Sprint-2		USN-11	As an editor, I can search, filter, sort and paginate FAQs.	3	Medium	Backend Dev 2
Sprint-2	AI Search & Answering	USN-12	As a developer, I can connect an external LLM and embedding API through a provider-agnostic AI service module with prompt templates.	3	High	AI Developer
Sprint-2		USN-13	As a developer, I want an embedding created and stored in pgvector whenever an FAQ is created or updated.	3	High	AI Developer
Sprint-2		USN-14	As a customer, I can ask a question and get the most relevant FAQs ranked by meaning.	3	High	AI Developer
Sprint-2	FAQ Management	USN-15	As an editor, I can bulk import FAQs from a CSV or JSON file.	3	Medium	Backend Dev 2
Sprint-3	AI Search & Answering	USN-16	As a customer, I can ask a question and receive a short AI-written answer based only on matching FAQs, with source links.	5	High	AI Developer
Sprint-3	Feedback & Escalation	USN-17	As a customer, I get a confidence score, and a clear "I don't know" reply with an option to reach a human when no FAQ matches well.	3	High	AI Developer
Sprint-3	AI Search & Answering	USN-18	As a customer, I can ask follow-up questions that use my earlier questions in the same session.	3	Medium	Backend Dev 2
Sprint-3	Feedback & Escalation	USN-19	As a customer, I can give a thumbs up or down on an answer.	2	Medium	Backend Dev 1
Sprint-3	Performance & Security	USN-20	As a product owner, I want repeated questions answered from a Redis cache to reduce cost and response time.	3	Medium	Backend Lead
Sprint-3		USN-21	As a product owner, I want rate limiting on question endpoints to prevent abuse and control cost.	2	High	Backend Lead

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Documentation	USN-22	As a developer, I can read interactive Swagger (OpenAPI) documentation for every endpoint.	2	High	Backend Lead
Sprint-4	Feedback & Escalation	USN-23	As a support agent, I can view and answer escalated and unanswered questions.	3	Medium	Backend Dev 1
Sprint-4	Analytics	USN-24	As an admin, I can see top questions, unanswered rate and satisfaction.	3	Medium	Backend Dev 2
Sprint-4	Account & Access	USN-25	As a third-party developer, I can use an API key to access the API from my own application.	3	High	Backend Lead
Sprint-4	Quality & Testing	USN-26	As a developer, I want unit tests for the auth, FAQ and AI service modules.	3	High	QA / Tester
Sprint-4		USN-27	As a developer, I want end-to-end tests covering the main API flows.	2	High	QA / Tester
Sprint-4		USN-28	As a developer, I want global validation, exception filters and logging for consistent error responses.	2	Medium	Backend Dev 1
Sprint-4	Deployment	USN-29	As a developer, I can build and test the project automatically through a CI/CD pipeline.	2	High	Backend Lead
Sprint-4		USN-30	As a product owner, I want the API deployed to a cloud environment with production configuration.	2	High	Backend Lead

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Feb 2025	01 Mar 2025		
Sprint-2	20	6 Days	03 Mar 2025	08 Mar 2025		
Sprint-3	20	6 Days	10 Mar 2025	15 Mar 2025		
Sprint-4	20	6 Days	17 Mar 2025	22 Mar 2025		
Total	80	24 Days	24 Feb 2025	22 Mar 2025		

Sprint dates run Monday to Saturday (6 days). Fill in the last two columns as each sprint finishes.

Velocity:

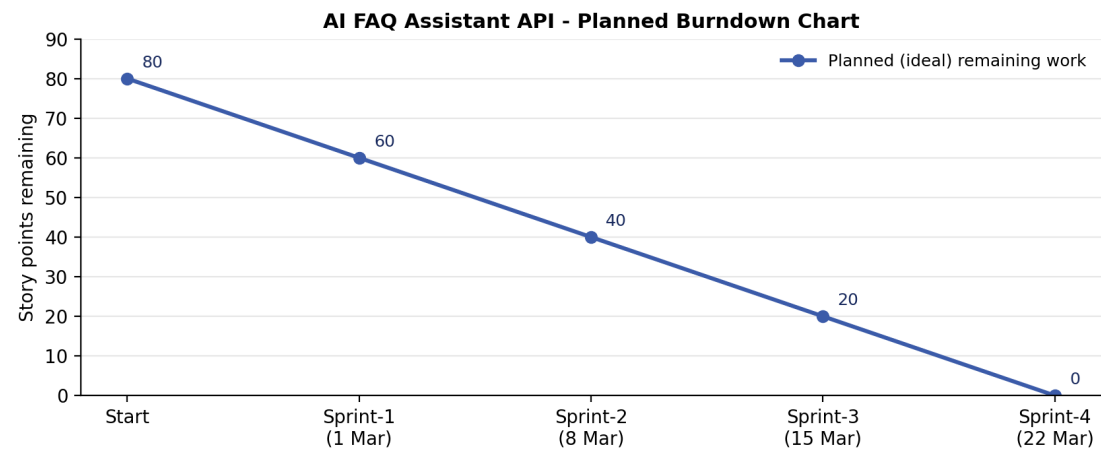
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day).

$$AV = velocity \div sprint\ duration = 20 \div 10 = 2\ story\ points\ per\ day$$

For AI FAQ Assistant API: velocity is 20 story points per sprint and each sprint lasts 6 days, so $AV = 20 \div 6 \approx 3.33$ story points per day.

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile <https://www.visual-paradigm.com/scrum/what-is-agile-software-development/> software development methodologies such as <https://www.visual-paradigm.com/scrum/scrum-in-3-minutes/> Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



The chart shows the planned (ideal) burndown. Record the actual points remaining at the end of each sprint in the table on the next page and add them to the chart as the project progresses.

Burndown data:

Sprint (end of)	Ideal points remaining	Actual points remaining
Start	80	
Sprint-1	60	
Sprint-2	40	
Sprint-3	20	
Sprint-4	0	

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>